Lab 23 — Distance-Weighted KNN

Aim

To implement **distance-weighted KNN** using inverse-distance weighting and compare its performance with uniform-weighted KNN.

Algorithm / Procedure

1. Load the Iris dataset.
2. Split the dataset into training and testing sets.
3. Create a uniform-weighted KNN classifier.
4. Create a distance-weighted KNN classifier.
5. Train both classifiers.
6. Predict the test data.
7. Calculate and compare their accuracies.

Program

```
from sklearn.datasets import load_iris

from sklearn.model_selection import train_test_split

from sklearn.neighbors import KNeighborsClassifier

from sklearn.metrics import accuracy_score

X,y=load_iris(return_X_y=True)

X_train,X_test,y_train,y_test=train_test_split(X,y,test_size=0.3,random_state=1)

uniform=KNeighborsClassifier(n_neighbors=5,weights='uniform')
```

```
distance=KNeighborsClassifier(n_neighbors=5,weights='distance')

uniform.fit(X_train,y_train)

distance.fit(X_train,y_train)

print("Uniform-weighted accuracy:",accuracy_score(y_test,uniform.predict(X_test)))

print("Distance-weighted accuracy:",accuracy_score(y_test,distance.predict(X_test)))
```

Output

Uniform-weighted accuracy: 0.9777777777777777

Distance-weighted accuracy: 0.9777777777777777

Result

The uniform-weighted and distance-weighted KNN classifiers were successfully implemented and compared. Both methods achieved approximately 97.8% accuracy on the Iris dataset.